



BIOINFORMATICS

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Topic 4

Determining a metagenomic sample composition

INTRODUCTION

Metagenomics involves the computational study of genetic material recovered directly from environmental, ecological, or clinical samples containing an uncultured mixture of multiple coexisting microorganisms. In traditional genomics, sequences are isolated from clonal cultures; however, environmental datasets present a highly complex mixture of fragmented DNA from hundreds of distinct species. A core challenge in processing these heterogeneous mixtures is taxonomic binning, which entails identifying the specific organism of origin for each raw genomic sequence (read) within the mixed sample. [1]

To evaluate this process under controlled conditions, this project implements a dual-method computational pipeline to classify long-read sequencing data generated via Pacific Biosciences (PacBio) technology across a set of distinct bacterial reference genomes. Utilizing high-throughput PacBio reads introduces distinct computational trade-offs: while their substantial length (often exceeding thousands of base pairs) provides critical structural context that spans across highly conserved genomic regions, these reads are characteristically prone to elevated rates of insertion and deletion (indel) sequencing errors. Consequently, developing robust computational frameworks is essential to accurately map these noisy sequences. By constructing a synthetic metagenome from a known selection of bacteria, this pipeline establishes a rigorous benchmark to directly analyze how these systematic errors affect classification accuracy and processing efficiency. [2]

The synthetic metagenome was constructed by combining PacBio sequencing reads originating from *Bacillus lentus*, *Vibrio mimicus*, *Escherichia coli*, *Salmonella enterica*, *Streptococcus infantis*, allowing direct validation of classification results against a known ground-truth composition.

To evaluate different approaches for taxonomic binning, this project compares an alignment-free method based on k-mer frequency distributions with an alignment-based baseline implemented using minimap2. This framework enables the assessment of how different computational strategies handle the error characteristics of PacBio long-read sequencing data. [3][4]

The aim of the project is to determine the composition of a synthetic metagenomic sample by assigning sequencing reads to their most likely bacterial reference genomes and evaluating the two methods in terms of classification accuracy and computational efficiency.

METHODOLOGY

Software Environment

The computational architecture of this project was implemented within **Google Colab**, a cloud-based Jupyter Notebook environment that provides a web-based platform for the execution of **Python source code**. By leveraging this cloud infrastructure, the development process utilized remote virtual machines to execute the sequencing data analysis, bypassing the constraints of local hardware configurations. The notebook-based structure of Google Colab allows a modular development approach, enabling system components—such as data parsing, k-mer profiling, and metric evaluation—to be executed and tested in isolated code cells while maintaining a continuous in-memory runtime session.

A major advantage of this environment is its **integration** with a robust ecosystem of pre-installed computational libraries and cloud storage management. The backend provides access to core scientific Python libraries such as **NumPy** and **Pandas** for optimized matrix manipulation, alongside the specialized BioPython framework, which serves as the foundational tool for parsing and manipulating raw genomic reads efficiently. Additionally, the platform facilitates a direct, authenticated integration with **Google Drive**. This connection acts as a persistent storage layer, allowing the pipeline to stream high-volume files, such as compressed sequencing datasets and reference bacterial genomes, directly into the active Python session without inducing heavy memory overhead.

Dataset

The dataset used in this project consisted of reference genomes and Pacific Biosciences (PacBio) long-read sequencing datasets obtained for five bacterial species: *Bacillus lentus*, *Vibrio mimicus*, *Salmonella enterica*, *Streptococcus infantis* and *Escherichia coli*. For each species, one reference genome in FASTA format and one sequencing dataset in FASTQ format were downloaded. These datasets were selected to provide a diverse collection of bacterial genomes for evaluating taxonomic classification methods

Metagenome Construction

A synthetic metagenomic dataset was generated by combining sequencing reads from five bacterial species: *Bacillus lentus*, *Escherichia coli*, *Salmonella enterica*, *Streptococcus infantis*, and *Vibrio mimicus*. For each species, 5,000 reads were extracted from the corresponding PacBio sequencing dataset and labelled using the format `species_name | read_id` to preserve their known origin.

The selected reads were merged into a single dataset and randomly shuffled to eliminate any ordering bias associated with the original sequencing files. The

resulting metagenome was stored in FASTA format and used as the input for both the k-mer based classifier and the minimap2 alignment-based approach.

The final synthetic metagenome contained 25,000 reads in total, corresponding to 5,000 reads from each bacterial species. The total size of the generated FASTA file was 99.60 MB.

Species	Reads	Mean read length (bp)	Approx. FASTA size (MB)
<i>Bacillus lentus</i>	5000	4457.91	21.71
<i>Escherichia coli</i>	5000	3637.63	17.80
<i>Salmonella enterica</i>	5000	3806.56	18.62
<i>Streptococcus infantis</i>	5000	5273.05	25.63
<i>Vibrio mimicus</i>	5000	2895.11	14.25
Total	25000	-	99.60

Classification Approaches

The project benchmarks an alignment-free heuristic approach against an alignment-based industry standard to evaluate classification accuracy and computational runtime performance:

- The Alignment-Free Pipeline: A custom Python implementation that profiles sequence similarity by mapping reads and references into an abstract vector space using normalized k-mer frequency distributions and Cosine Similarity optimization.
- The Alignment-Based Baseline: An explicit sequence alignment pipeline executed via minimap2 using the specific parameter preset -x map-pb. This preset optimizes the affine gap-scoring matrix to tolerate the high insertion/deletion (indel) error rates characteristic of raw PacBio long reads. The resulting PAF alignments served as the verification baseline.

Alignment-Free Classification using k-mers

The alignment-free classification method was implemented using normalized k-mer frequency distributions. First, each reference genome was indexed by extracting all valid k-mers of length *k* containing only the standard DNA bases A, C, G, and T.

The raw k-mer counts were then normalized by dividing each count by the total number of valid k-mers, producing a frequency profile for each reference genome.

The same procedure was applied to every read in the synthetic metagenomic sample. For each read, k-mer profiles were computed for both the forward strand and the reverse complement, since sequencing reads may align in either orientation. Each read profile was then compared against all reference genome profiles using cosine similarity:

$$S(V_R, V_G) = \frac{V_R \cdot V_G}{\|V_R\| \|V_G\|}$$

where V_R represents the k-mer frequency vector of the read, V_G represents the k-mer frequency vector of the reference genome, $V_R \cdot V_G$ is the dot product between the two vectors, and $\|V_R\|$ and $\|V_G\|$ denote their Euclidean norms.

Cosine similarity measures the angle between two frequency vectors and produces a value between 0 and 1, where values closer to 1 indicate a higher degree of similarity in k-mer composition. Unlike simple count comparisons, this metric focuses on the relative distribution of k-mers rather than the absolute sequence length, making it suitable for comparing short reads against complete bacterial genomes. [5]

The reference genome with the highest similarity score was selected as the predicted organism of origin. Because the true species label had been stored in the read identifier during metagenome construction, the predicted label could be directly compared with the real label. This allowed the calculation of the total number of correctly classified reads, the overall classification accuracy, and the predicted metagenomic composition for each bacterial genome.

Example of read classification using K-mers

Step 1: Read representation

Consider the read: ATATAT

Using $k = 2$: AT, TA, AT, TA, AT

k-mer	Count	Frequency
AT	3	3/5=0.60
TA	2	2/5=0.40

Read vector: $V_R = [AT = 0.60, TA = 0.40]$

Step 2: Reference Genome A

Sequence: ATAT

Using k = 2: AT, TA, AT

k-mer	Count	Frequency
AT	2	2/3=0.66
TA	1	1/3=0.33

Reference vector: $V_{GA} = [AT = 0.66, TA = 0.33]$

Step 3: Reference Genome B

Sequence: GGGG

Using k = 2: GG, GG, GG

k-mer	Count	Frequency
GG	3	3/3=1.00

Reference vector: $V_{GB} = [GG = 1.00]$

Step 4: Cosine Similarity Calculation

Similarity formula:

$$S(V_R, V_G) = \frac{V_R \cdot V_G}{\|V_R\| \|V_G\|}$$

Norm of the read: $\|V_R\| = \sqrt{0.6^2 + 0.4^2} = 0.72$

Genome A:

Norm of Genome A: $\|V_{GA}\| = \sqrt{0.66^2 + 0.33^2} = 0.74$

Dot product: $0.6 \cdot 0.66 + 0.4 \cdot 0.33 = 0.528$

Cosine similarity: $S(V_R, V_{GA}) = \frac{0.528}{0.72 \cdot 0.74} \approx 0.99$

Genome B:

$$\|V_{GB}\| = \sqrt{1.00^2} = 1$$

Norm of Genome B:

Dot product: $V_R \cdot V_{GB} = 0$

Cosine similarity:

$$S(V_R, V_{GB}) = \frac{0}{\|V_R\| \cdot \|V_{GB}\|} = 0$$

Cosine similarity is 0 because no k-mers are shared.

Step 5: Classification Decision

Since:

- Similarity (Read, Genome A) = 0.99
- Similarity (Read, Genom B) = 0

The read is assigned to Genome A.

Alignment-Based Classification using minimap2

After the preliminary test, minimap2 was used as the alignment-based baseline for classifying reads in the synthetic metagenomic sample. The complete metagenome file was aligned independently against each bacterial reference genome using the PacBio-specific preset `-x map-pb`. For each reference genome, minimap2 generated a PAF output file containing the alignments between the metagenomic reads and that reference.

The PAF (Pairwise Alignment Format) file generated by minimap2 stores alignment information in a tab-separated format. The first twelve columns are mandatory and contain essential information about the read, the reference genome, and the alignment itself. In this project, four fields were extracted from each PAF record: the read identifier (column 1), the number of matching bases (column 10), the alignment block length (column 11), and the mapping quality score, MAPQ (column 12). Since Python indexing starts at 0, these PAF columns correspond to **fields[0]**, **fields[9]**, **fields[10]**, and **fields[11]** in the implementation. These values were used to estimate alignment quality and determine the most likely reference genome for each read.

PAF Column	Variable	Description
1	read_id	Sequencing read identifier
10	matching_bases	Number of matching bases in the alignment

11	alignment_length	Total alignment block length
12	MAPQ	Mapping quality score

An alignment identity value was calculated as the ratio between matching bases and alignment length. To select the best reference for each read, a classification score was defined as the product of alignment identity and mapping quality (MAPQ). If a read aligned to multiple references, the reference with the highest score was selected as the predicted organism of origin.

The predicted labels obtained from minimap2 were then compared with the true species labels stored in the read identifiers. These labels had been embedded during metagenome construction using the format **species_name | read_id**, allowing direct evaluation of classification accuracy. This allowed the calculation of the number of reads producing valid alignments, the number of correctly classified reads, the minimap2 classification accuracy, and the estimated metagenomic composition. Finally, the minimap2 predictions were compared with the k-mer based predictions to calculate the agreement between both methods.

Example of minimap2 classification

Consider a read that produces the following alignment results:

Reference genome	Matching bases	Alignment length	MAPQ
Genome A	870	900	60
Genome B	500	850	20

The alignment identity is calculated as:

- Identity = Matching Bases / Alignment Length

For Genome A:

$$\text{Identity_A} = 870 / 900 = 0.967$$

$$\text{Score_A} = \text{Identity_A} \times \text{MAPQ_A}$$

$$\text{Score_A} = 0.967 \times 60 = 58.0$$

For Genome B:

$$\text{Identity_B} = 500 / 850 = 0.588$$

$$\text{Score_B} = \text{Identity_B} \times \text{MAPQ_B}$$

$$\text{Score_B} = 0.588 \times 20 = 11.8$$

Classification Decision

Since $\text{Score_A} > \text{Score_B}$ ($58.0 > 11.8$) the read is assigned to Genome A.

This example directly reflects the implementation used in the project. For each alignment reported by minimap2, the values **matching_bases**, **alignment_length**, and **mapq** are extracted from the PAF file. The alignment identity is computed as the ratio between matching bases and alignment length, while the final classification score used in this project is calculated as the product of identity and MAPQ.

In the implementation, these operations correspond to:

- `matching_bases = int(fields[9])`
- `alignment_length = int(fields[10])`
- `mapq = int(fields[11])`
- `identity = matching_bases / alignment_length`
- `score = identity * mapq`

If a read aligns to multiple reference genomes, only the alignment with the highest score is retained and used as the final prediction. This is achieved by comparing the score of each alignment and keeping the reference genome that produces the maximum value. The reference with the highest score is therefore selected as the predicted organism of origin.

RESULTS

K-mer classification

The experimental pipeline was evaluated across an odd-integer sweep of k values ($k = 5, 7, 9, 11, 13, 15, 17, 19, 21, 23$) using a synthetic metagenomic sample containing 5,000 reads per bacterium. The metrics recorded are summarized below:

	Accuracy	Execution time
k = 5	79.18%	3 min 11 s
k = 7	80.80%	8 min 13 s
k = 9	83.19%	25 min 18 s
k = 11	85.87%	23 min 2 s
k = 13	91.12%	17 min 13 s

k = 15	93.89%	18 min 36 s
k = 17	94.43%	19 min 37 s
k = 19	94.46%	20 min 21 s
k = 21	93.72%	21 min 3 s
k = 23	92.42%	21 min 10 s

Classification accuracy increased progressively with larger k-mer sizes, reaching a maximum value of 94.46% at k = 19. Beyond this point, accuracy decreased for k = 21 and k = 23, suggesting that larger k-mers became more sensitive to sequencing errors and produced increasingly sparse frequency profiles. Therefore, k = 19 was selected as the optimal parameter for subsequent analyses.

Minimap2 classification

Minimap2 was used as an alignment-based baseline for taxonomic classification. Out of the 25,000 reads contained in the synthetic metagenome, 22,394 produced at least one valid alignment against the reference genomes, corresponding to a coverage of 89.58%.

Among the aligned reads, 22,116 were correctly classified, resulting in an accuracy of 98.76% on aligned reads. When the complete metagenome was considered, including reads without reported alignments, the global classification accuracy was 88.46%.

Metric	Value
Total reads in metagenome	25000
Read with valid alignments	22394
Coverage	89.58%
Correctly classified aligned reads	22116
Accuracy on aligned reads	98.76%
Global accuracy (all reads)	88.46%
Execution time	45.93 s

Comparison Between k-mer and minimap2 classification

The predictions generated by the k-mer classifier were compared with those obtained using Minimap2. Since Minimap2 only produced classifications for reads with at least one valid alignment, the comparison was restricted to the subset of 22,394 reads classified by both methods.

Metric	Value
Common reads compared	22394
Same predictions	22039
Agreement	98.41%
K-mer accuracy on common reads	99.12%
Minimap2 accuracy on common reads	98.76%

Agreement was calculated as the percentage of reads for which both methods produced the same taxonomic assignment. This metric reflects the consistency between the two classification approaches and should not be interpreted as classification accuracy.

To further examine the classification performance of both methods, confusion matrices were generated using the common subset of reads:

CONFUSION MATRIX FOR THE K-MER CLASSIFIER

	<i>bacillus_lentus</i>	<i>escherichia_coli</i>	<i>salmonella_enterica</i>	<i>streptococcus_infantis</i>	<i>vibrio_mimicus</i>
<i>bacillus_lentus</i>	4708	5	0	4	1
<i>escherichia_coli</i>	5	4313	45	2	0
<i>salmonella_enterica</i>	21	64	4074	14	5
<i>streptococcus_infantis</i>	3	5	2	4713	3
<i>vibrio_mimicus</i>	3	8	3	4	4389

CONFUSION MATRIX FOR MINIMAP2

	<i>bacillus_lentus</i>	<i>escherichia_coli</i>	<i>salmonella_enterica</i>	<i>streptococcus_infantis</i>	<i>vibrio_mimicus</i>
<i>bacillus_lentus</i>	4711	0	1	6	0
<i>escherichia_coli</i>	0	4255	108	1	1
<i>salmonella_enterica</i>	2	108	4066	0	2
<i>streptococcus_infantis</i>	2	0	7	4717	0
<i>vibrio_mimicus</i>	2	29	3	6	4367

The confusion matrices revealed high classification performance for both methods, with most misclassifications occurring between *Escherichia coli* and *Salmonella enterica*.

DISCUSSION

The evaluation of different k-mer sizes showed a clear relationship between k and classification performance. Accuracy increased from 79.18% at k = 5 to a maximum of 94.46% at k = 19, after which performance declined slightly. This trend suggests that intermediate k-mer sizes provide the best balance between taxonomic specificity and robustness to sequencing errors. Smaller k values are less informative because many k-mers are shared among different genomes, whereas excessively large k values become more sensitive to sequencing errors and generate increasingly sparse frequency profiles.

Minimap2 was used as an alignment-based baseline and achieved an accuracy of 98.76% on aligned reads. However, valid alignments were obtained for only 89.58% of the reads in the synthetic metagenome. When reads without valid alignments were treated as classification errors, the global classification accuracy decreased to 88.46%. This highlights an important characteristic of alignment-based approaches: although they can achieve high classification accuracy, reads that fail to produce sufficiently reliable alignments remain unclassified.

The comparison between both methods showed a high degree of consistency. On the subset of 22,394 reads classified by both approaches, the agreement reached 98.41%, indicating that the k-mer classifier and Minimap2 produced highly similar taxonomic assignments for the vast majority of reads. Furthermore, both methods achieved very similar classification performance on this common subset, with accuracies of 99.12% and 98.76% for the k-mer classifier and Minimap2, respectively.

The confusion matrices further demonstrated that most classification errors were concentrated between *Escherichia coli* and *Salmonella enterica*. This observation is expected given the close evolutionary relationship between these species and the substantial sequence similarity shared across many genomic regions. Misclassifications involving the remaining species were comparatively rare.

Regarding computational performance, Minimap2 completed the classification in 45.93 seconds, whereas the k-mer classifier required substantially longer execution time (20 min 21s), approximately 27 times longer than Minimap2. Although alignment-free methods are frequently described as computationally efficient, the implementation developed in this project performs an exhaustive comparison between read and reference k-mer frequency profiles using cosine similarity. In

contrast, Minimap2 relies on highly optimized indexing and seeding algorithms, which explains its considerably faster execution despite performing sequence alignments.

Memory consumption was not measured directly during the experiments. Nevertheless, the k-mer implementation stores only observed k-mers using sparse dictionary structures rather than allocating the complete theoretical k-mer space (4^k). This design reduces memory requirements and allows larger k values to be evaluated without excessive memory usage. Future work could incorporate explicit memory profiling to provide a more complete comparison between both approaches.

Overall, the results demonstrate that both approaches are capable of accurately estimating metagenomic composition. While Minimap2 showed a clear advantage in execution time, both methods produced highly consistent taxonomic assignments across the synthetic metagenome.

CONCLUSION

This project successfully implemented and evaluated two complementary approaches for taxonomic classification of a synthetic metagenomic sample composed of five bacterial species sequenced with PacBio long-read technology.

The alignment-free k-mer classifier demonstrated a clear improvement in accuracy as k increased, reaching a peak of **94.46% at k = 19**, confirming that longer k-mers capture more species-specific sequence signatures. The alignment-based minimap2 baseline achieved an accuracy of **98.76%** on aligned reads and **88.46%** globally, with valid alignments obtained for **89.58%** of the total reads. Despite the inherent difference in methodology, the two approaches showed a high agreement of **98.41%** on the common subset of classified reads, indicating that the k-mer classifier is capable of producing taxonomic assignments largely consistent with those obtained through sequence alignment.

These results suggest that alignment-free k-mer profiling combined with cosine similarity represents a viable and competitive strategy for metagenomic classification, capable of classifying **all reads** without relying on sequence alignment. The pipeline developed in this project provides a solid foundation for future work on metagenome composition analysis.

LIMITATIONS

Several limitations should be acknowledged when interpreting the results of this project. First, the large volume of data processed—25,000 long PacBio reads across five reference genomes—resulted in substantial execution times, particularly for the k-mer classifier at higher values of k. Running the full parameter sweep required several hours of computation, which constrained the ability to perform a more exhaustive exploration of k values or to repeat experiments multiple times for statistical validation.

Second, the high memory demands of storing k-mer frequency profiles for large reference genomes and thousands of reads simultaneously limited the scalability of the pipeline; for larger metagenomic datasets, the current in-memory implementation would likely require architectural changes such as on-disk indexing or distributed processing.

Third, memory consumption was not formally profiled during the experiments, meaning the exact memory footprint of the pipeline at each k value remains unquantified.

Fourth, the comparison between the k-mer classifier and minimap2 was limited to the subset of reads for which minimap2 produced a valid alignment; the 10.42% of unaligned reads were excluded from the agreement analysis, which may introduce a slight bias towards overestimating concordance between the two methods.

Finally, the synthetic metagenome used in this study was composed of only five bacterial species. While this controlled setup allowed for precise evaluation of both classification methods, real metagenomic samples typically contain hundreds to thousands of distinct microbial species, many of which may be absent from reference databases. The limited species diversity of the dataset may therefore inflate the observed classification accuracies, as both methods were only required to distinguish between a small number of reference genomes. Performance on more complex, real-world metagenomic samples could differ substantially.

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